

Concept 2 | Exponential Functions and Their Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Exponential

$$y = ab^x$$

Growth

$$A = A_0(1 + r)^t$$

Decay

$$A = A_0(1 - r)^t$$

Worked Solutions

Exponential Functions and Their Applications

Q001 Written

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

Solution

1. Plot $x = -1, 0, 1$. A base > 1 gives growth; a base between 0 and 1 gives decay. The curve stays above the x-axis.

Final answer

Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.

Q002 Written

Draw $y = 2^x$ and $y = (1/2)^x$.

Solution

1. Use $(-1, 1/2), (0, 1), (1, 2)$ for 2^x and $(-1, 2), (0, 1), (1, 1/2)$ for its reciprocal-base graph.

Final answer

2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.

Worked Solutions

Exponential Functions and Their Applications

Q003 Written

Draw the six source graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

Solution

1. For each graph mark $x = -1, 0, 1$, classify the base, and sketch a curve that never meets the x-axis.

Final answer

Bases 4, 10, 3/2: strictly increasing. Bases 2/3, 1/4, 1/10: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.

Q004 Written

Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.

Solution

1. Because $\pi > 1$, the curve increases. Restrict the drawing to the interval $[-1, 2]$ and keep $y > 0$.

Final answer

Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.

Worked Solutions

Exponential Functions and Their Applications

Q005 Written

Order the source sets 0.7^{-3} , 0.7^{-1} , 0.7^2 and $5/8$, $5/64$, $6/16$ using inequality signs.

Solution

1. For a base $0 < a < 1$, larger exponents give smaller values. For the second set rewrite with a common base 2 and compare exponents.

Final answer

$$0.7^2 < 0.7^{-1} < 0.7^{-3}; \text{ and } 5/64 < 5/8 < 6/16.$$

Q006 Written

Order the three displayed Try sets: 5^0 , 5^{-4} , $5^{3/2}$; $(1/3)^7$, $(1/3)^2$, $(1/3)^{-5}$; and 3^0 , 3^2 , 3^{-2} .

Solution

1. Make the base common. Keep exponent order for base > 1 and reverse it for $0 < a < 1$.

Final answer

$$5^{-4} < 5^0 < 5^{3/2}; (1/3)^7 < (1/3)^2 < (1/3)^{-5}; 3^{-2} < 3^0 < 3^2.$$

Worked Solutions**Exponential Functions and Their Applications****Q007** Written**Order the source sets with bases 3, 2, $\frac{3}{2}$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.****Solution**

1. The nine source comparisons are solved by one rule: rewrite, state whether the base is above or below one, then write the ordered chain.

Final answer

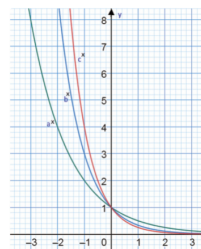
Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.

Worked Solutions

Exponential Functions and Their Applications

Q008 Written

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a , b , c .



Solution

1. Convert radicals to rational powers before ordering; the opposite graph reverses the order when its base lies between 0 and 1.

Final answer

Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.

Worked Solutions

Exponential Functions and Their Applications

Q009 Written

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

Solution

1. Rewrite with a common base > 1 , compare exponents, and reverse an inequality only when the chosen base is between 0 and 1.

Final answer

$x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.

Q010 Written

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, **and the reciprocal-base inequalities.**

Solution

1. The Try set is source-faithfully solved by the same two-step method: common base, then exponent equation/inequality with sign control.

Final answer

Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .

Worked Solutions

Exponential Functions and Their Applications

Q011 Written

Solve all 24 source equations and inequalities in Exercise (a)–(x), including the 2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125 common-base forms.

Solution

1. For each subpart: factor numerical constants into one base, write the exponent comparison, state $\text{base} > 1$ or $0 < \text{base} < 1$, and solve.

Final answer

Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.

Q012 Written

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

Solution

1. Use base $4 > 1$, solve the quadratic inequality, then apply the physical time restriction $x \geq 0$.

Final answer

$x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.

Worked Solutions

Exponential Functions and Their Applications

Q013 Written

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

Solution

1. Set $t = a^x > 0$, solve the quadratic in t , discard non-positive roots, then convert back to x .

Final answer

First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.

Q014 Written

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

Solution

1. The substitution changes the exponential problem to a quadratic; the positivity condition on t is mandatory before returning to x .

Final answer

Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.

Worked Solutions

Exponential Functions and Their Applications

Q015 Written

Solve the twelve source items (a)–(l), each reducible to a quadratic in 2^x or 3^x .

Solution

1. Write $t > 0$ beside the substitution, solve the quadratic, reject $t \leq 0$, and only then use monotonicity of the base to recover x .

Final answer

For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .

Q016 Written

Solve completely for real x : $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0$.

Solution

1. Factor the quadratic as $(t - 1)(t - 4) \leq 0$, convert the bounds with the decreasing base $1/2$, then solve the resulting quadratic inequalities.

Final answer

Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.

Worked Solutions

Exponential Functions and Their Applications

Q017 Written

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x -values.

Solution

1. Find the exact t -range first, compare the vertex with both endpoints, then convert t back to x .

Final answer

Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.

Q018 Written

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x -values.

Solution

1. The quadratic has vertex $t = -1$, outside the positive t -range, so compare the two endpoints.

Final answer

Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.

Worked Solutions

Exponential Functions and Their Applications

Q019 Written

Solve the three source extrema models: $-3 \leq x \leq 2$, $y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3$, $y = -4 + 2^{2x} - 3^{x+2}$; **and the third displayed interval/model.**

Solution

1. A correct solution must include the t-range; an unconstrained quadratic vertex is not accepted unless it lies in that range.

Final answer

In each case set $t = a^x$, write the exact t-range, and compare the quadratic vertex and endpoints; report both extreme values and their x-locations.

Q020 Written

For $-1 \leq x \leq 2$, **optimise** $y = 2 \cdot 4^x - 2^{x+2} + 11$.

Solution

1. Use the interval $1/2 \leq t \leq 4$, evaluate the vertex $t = 1$, and compare both endpoints.

Final answer

Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.

Worked Solutions

Exponential Functions and Their Applications

Q021 **Written**

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

Solution

1. AM-GM gives $a^x + a^{-x} \geq 2$, with equality at $a^x = 1$, i.e. $x = 0$; the two terms move in opposite directions away from zero.

Final answer

h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.

Q022 **MCQ**

Which base gives a strictly decreasing exponential graph?

Solution

1. A base between 0 and 1 gives decay.

Correct option: C

Final answer

$1/2$

Worked Solutions

Exponential Functions and Their Applications

Q023 MCQ

The horizontal asymptote of $y = a^x$ is:**Solution**

1. Exponential values stay positive and approach zero.

Correct option: B**Final answer**

$$y = 0$$

Q024 MCQ

The range of $y = 5^x$ is:**Solution**

1. Every exponential output is positive.

Correct option: C**Final answer**

$$(0, \infty)$$

Worked Solutions

Exponential Functions and Their Applications

Q025 MCQ

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?**Solution**1. At $x = 1$, $y = \pi$.

Correct option: B

Final answer $(1, \pi)$

Q026 MCQ

For $0 < a < 1$ and $r < s$, which is true?**Solution**

1. The exponent order reverses below base 1.

Correct option: B

Final answer $a^r > a^s$

Worked Solutions

Exponential Functions and Their Applications

Q027 MCQ

The first step when bases differ is to:**Solution**

1. Rewrite all terms using one base.

Correct option: B**Final answer**

make the bases equal

Q028 MCQ

Which is largest? 3^{-2} , 3^0 , 3^2 **Solution**

1. Base 3 is greater than 1.

Correct option: C**Final answer** 3^2

Worked Solutions

Exponential Functions and Their Applications

Q029 MCQ

The challenge term $(\sqrt[3]{4})^3$ equals:

Solution

1. $(2^{2/3})^3 = 2^2$.

Correct option: B

Final answer

$$2^2$$

Q030 MCQ

Solving $4^{x-1} = 8^{x+3}$ gives:

Solution

1. Use $2^{2x-2} = 2^{3x+9}$.

Correct option: A

Final answer

$$-11$$

Worked Solutions

Exponential Functions and Their Applications

Q031 MCQ

For $0 < a < 1$, an exponential inequality requires:**Solution**

1. Decreasing functions reverse inequalities.

Correct option: B**Final answer**

a reversed exponent order

Q032 MCQ

The solution of $3^x = 81$ is:**Solution**

1. $81 = 3^4$.

Correct option: C**Final answer**

4

Worked Solutions

Exponential Functions and Their Applications

Q033 MCQ

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

Solution

1. Solve $(x - 2)(x + 1) > 0$, then use $x \geq 0$.

Correct option: B

Final answer

$$x > 2$$

Q034 MCQ

In substitution methods, $t = a^x$ must satisfy:

Solution

1. Every positive base power is positive.

Correct option: C

Final answer

$$t > 0$$

Worked Solutions

Exponential Functions and Their Applications

Q035 MCQ

The positive root of $t^2 - 3t - 16 = 0$ is:**Solution**1. Discard the negative root because $t > 0$.**Correct option:** B**Final answer**

$$(3 + \sqrt{73})/2$$

Q036 MCQ

The challenge quadratic in t is:**Solution**1. Set $t = (1/2)^{x^2 - 2x}$.**Correct option:** B**Final answer**

$$t^2 - 5t + 4$$

Worked Solutions

Exponential Functions and Their Applications

Q037 MCQ

After solving an exponential quadratic, reject roots with:**Solution**

1. A power a^x cannot be zero or negative.

Correct option: C**Final answer**

$$t \leq 0$$

Q038 MCQ

For extrema, the first required range is the range of:**Solution**

1. The quadratic is constrained by the exact t-range.

Correct option: B**Final answer**

$$t = a^x$$

Worked Solutions

Exponential Functions and Their Applications

Q039 MCQ

The minimum of $y = t^2 - 4t$ at $t = 2$ is:**Solution**1. Complete the square: $(t - 2)^2 - 4$.

Correct option: B

Final answer

-4

Q040 MCQ

The C06-K02 challenge has minimum value:**Solution**1. $y = 2(t - 1)^2 + 9$.

Correct option: A

Final answer $9atx = 0$

Worked Solutions

Exponential Functions and Their Applications

Q041 MCQ

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:

Solution

1. AM–GM gives a minimum of 2 at $x = 0$.

Correct option: C

Final answer

2

Q042 MCQ

At which value of x does $a^x + a^{-x}$ attain its minimum?

Solution

1. Equality in AM–GM occurs when $a^x = 1$.

Correct option: B

Final answer

$x = 0$

Worked Solutions

Exponential Functions and Their Applications

Quiz MCQ 1 **Concept Quiz · MCQ**

Which statement about $y = a^x$ is always true for $a > 0$, $a \neq 1$?

Solution

1. The domain is all real numbers and the range is positive.

Correct option: A

Final answer

range $(0, \infty)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For base 0.5, increasing the exponent makes the value:

Solution

1. The function is decreasing.

Correct option: B

Final answer

smaller

Worked Solutions

Exponential Functions and Their Applications

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The positive-substitution condition is:

Solution

1. An exponential power is always positive.

Correct option: C

Final answer

$$t > 0$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

For $y = 2(t - 1)^2 + 9$, the minimum is:

Solution

1. The square is zero at $t = 1$.

Correct option: C

Final answer

$$9$$

Worked Solutions

Exponential Functions and Their Applications

Quiz WRITTEN 5 **Concept Quiz · Written**

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

Solution

1. Use the three points and the base $0 < 1/4 < 1$.

Final answer

Decreasing; $D = \mathbb{R}, R = (0, \infty), y = 0$

Quiz WRITTEN 6 **Concept Quiz · Written**

Order $2^{-3}, 2^0, 2^{5/2}$.

Solution

1. Base 2 is greater than 1, so preserve exponent order.

Final answer

$$2^{-3} < 2^0 < 2^{5/2}$$

Worked Solutions

Exponential Functions and Their Applications

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $4^x - 5 \cdot 2^x + 4 = 0$.**Solution**1. Let $t = 2^x > 0$: $t^2 - 5t + 4 = (t - 1)(t - 4) = 0$.**Final answer**

$$x = 0 \text{ or } x = 2$$

Quiz WRITTEN 8 Concept Quiz · Written**Find the minimum of** $a^x + a^{-x}$ **for** $a > 1$.**Solution**1. By AM-GM, $a^x + a^{-x} \geq 2$, with equality when $a^x = 1$.**Final answer**

$$2 \text{ at } x = 0$$

Answer key

Use the question number shown on each page.

- Q001** Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.
- Q002** 2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.
- Q003** Bases 4, 10, $3/2$: strictly increasing. Bases $2/3$, $1/4$, $1/10$: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.
- Q004** Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.
- Q005** $0.7^2 < 0.7^{-1} < 0.7^{-3}$; and $5/64 < 5/8 < 6/16$.
- Q006** $5^{-4} < 5^0 < 5^{3/2}$; $(1/3)^7 < (1/3)^2 < (1/3)^{-5}$; $3^{-2} < 3^0 < 3^2$.
- Q007** Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.
- Q008** Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.
- Q009** $x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.
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- Q012** $x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.
- Q013** First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.
- Q014** Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.
- Q015** For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .
- Q016** Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.
- Q017** Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.
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- Q019** In each case set $t = a^x$, write the exact t -range, and compare the quadratic vertex and endpoints; report both extreme values and their x -locations.
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- Q021** h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.
- Q022** C
- Q023** B
- Q024** C
- Q025** B
- Q026** B
- Q027** B
- Q028** C
- Q029** B
- Q030** A
- Q031** B
- Q032** C
- Q033** B
- Q034** C
- Q035** B
- Q036** B
- Q037** C
- Q038** B
- Q039** B
- Q040** A
- Q041** C
- Q042** B
- Quiz MCQ 1** A
- Quiz MCQ 2** B
- Quiz MCQ 3** C
- Quiz MCQ 4** C
- Quiz WRITTEN 5** Decreasing; $D = \mathbb{R}, R = (0, \infty), y =$
- Quiz WRITTEN 6** $2^{-3} < 2^0 < 2^{5/2}$
- Quiz WRITTEN 7** $x = 0$ or $x = 2$
- Quiz WRITTEN 8** $2atx = 0$